

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. IV YEAR**  
**INFORMATION TECHNOLOGY**

**VII SEMESTER**

**R22**

| Course Code | Title of the Course                             | L  | T | P/D | CH | C  |
|-------------|-------------------------------------------------|----|---|-----|----|----|
| 22PC1DS303  | Big Data Computing Technologies                 | 3  | 0 | 0   | 3  | 3  |
| 22PC1DS402  | Deep Learning Applications                      | 3  | 0 | 0   | 3  | 3  |
|             | PROFESSIONAL ELECTIVE – III                     |    |   |     |    |    |
| 22PE1IN303  | Adhoc and Sensor Networks                       | 3  | 0 | 0   | 3  | 3  |
| 22PE1IT401  | Soft Computing                                  |    |   |     |    |    |
| 22PE1DS402  | Distributed Computing                           |    |   |     |    |    |
| 22PE1IT402  | Software Project Management                     |    |   |     |    |    |
| 22PE1CS401  | Advanced Databases                              |    |   |     |    |    |
|             | PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL |    |   |     |    |    |
|             | Artificial Intelligence Stream                  | 3  | 0 | 0   | 3  | 3  |
|             | Data Science Stream                             |    |   |     |    |    |
|             | Cyber Security Stream                           |    |   |     |    |    |
|             | Artificial Intelligence Stream                  |    |   |     |    |    |
|             | Machine Learning Stream                         |    |   |     |    |    |
|             | OPEN ELECTIVE – III                             | 3  | 0 | 0   | 3  | 3  |
| 22PC2DS303  | Big Data Computing Technologies Laboratory      | 0  | 0 | 2   | 2  | 1  |
| 22PC2DS402  | Deep Learning Applications Laboratory           | 0  | 0 | 2   | 2  | 1  |
| 22PW4IT401  | Major Project Phase – I                         | 0  | 0 | 6   | 6  | 3  |
| Total       |                                                 | 15 | 0 | 10  | 25 | 20 |

L – Lecture      T – Tutorial      P – Practical      D – Drawing  
C – Credits      SE – Sessional Examination      CA – Class Assessment  
SEE – Semester End Examination      D-D – Day to Day Evaluation  
CP – Course Project      PE – Practical Examination

CH – Contact Hours/Week  
ELA – Experiential Learning Assessment  
LR – Lab Record

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. IV YEAR**  
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**VIII SEMESTER**

**R22**

| Course Code  | Title of the Course                   | L        | T        | P/D       | CH        | C         |
|--------------|---------------------------------------|----------|----------|-----------|-----------|-----------|
|              | <b>PROFESSIONAL ELECTIVE – V</b>      |          |          |           |           |           |
| 22PE1IT403   | Adversarial Machine Learning          | 3        | 0        | 0         | 3         | 3         |
| 22PE1CS403   | Computer Vision                       |          |          |           |           |           |
| 22PE1AM405   | Cognitive Engineering                 |          |          |           |           |           |
| 22PE1CS404   | Augmented Reality and Virtual Reality |          |          |           |           |           |
| 22PC1CY401   | Block Chain Technology                |          |          |           |           |           |
|              | <b>PROFESSIONAL ELECTIVE – VI</b>     |          |          |           |           |           |
| 22PC1AM304   | Natural Language Processing           | 3        | 0        | 0         | 3         | 3         |
| 22PE1CS405   | Social Media Analytics                |          |          |           |           |           |
| 22PE1DS405   | Time Series Analysis and Forecasting  |          |          |           |           |           |
| 22PE1AM302   | Explainable AI                        |          |          |           |           |           |
| 22PE1DS406   | Quantum Computing                     |          |          |           |           |           |
|              | <b>OPEN ELECTIVE – IV</b>             | 3        | 0        | 0         | 3         | 3         |
| 22PW4IT402   | Major Project Phase – II              | 0        | 0        | 22        | 22        | 11        |
| <b>Total</b> |                                       | <b>9</b> | <b>0</b> | <b>22</b> | <b>31</b> | <b>20</b> |

L – Lecture      T – Tutorial      P – Practical      D – Drawing  
C – Credits      SE – Sessional Examination      CA – Class Assessment  
SEE – Semester End Examination      D-D – Day to Day Evaluation  
CP – Course Project      PE – Practical Examination

CH – Contact Hours/Week  
ELA – Experiential Learning Assessment  
LR – Lab Record

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PC1DS303) BIG DATA COMPUTING TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To explore the fundamental concepts of data analytics
- To learn to analyze the data analysis techniques
- To explore the techniques related to mining streams
- To understand, explore Hadoop framework technology and its associated tools
- To apply data visualization methods

**COURSE OUTCOMES:** After completion of the course, the student should be able to  
**CO-1:** Identify the major sources of data; apply pre-processing tasks and summary statistics on data

**CO-2:** Understand the concept of Hadoop and analyses data on HDFS and its comparison with other systems

**CO-3:** Work with Hadoop framework such as PIG, HIVE

**CO-4:** Explore various Hadoop components such as HBASE, ZOOKEEPER etc.

**CO-5:** Build various models and visualize data using visualization tools for gaining more insights

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 2                     | 2    | 2    | 2    | 3    | 2    | 1    | 2    | 2    | 2     | 2     | 1     | 2                               | 1     | 2     | 1     |
| CO-2 | 2                     | 3    | 2    | 2    | 3    | 2    | 1    | 1    | 2    | 2     | 2     | 2     | 3                               | 2     | 3     | 1     |
| CO-3 | 2                     | 3    | 2    | 2    | 3    | 1    | 1    | 3    | 2    | 2     | 2     | 2     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 2    | 3    | 3    | 2    | 1    | 2    | 2    | 2     | 2     | 2     | 2                               | 3     | 2     | 2     |
| CO-5 | 2                     | 3    | 2    | 3    | 3    | 1    | 1    | 1    | 2    | 2     | 2     | 2     | 3                               | 2     | 3     | 2     |

#### UNIT-I:

**Data Management:** Introduction to Big Data, Big Data Enabling Technologies, Sources of data, Types of Data, Data preprocessing, Data collection and APIs, Exploring and fixing data, Data storage and management, using multiple data sources. Data Analysis: Introduction, Terminology and concepts, Summary statistics, Central tendencies and distributions, Variance.

#### UNIT-II:

**Hadoop:** Meet Hadoop, Hadoop Stack for Big Data, A brief history of Hadoop and the Hadoop ecosystem, Analyzing the Data with Hadoop, Hadoop Distributed File System, HDFS concepts, Design of HDFS, Data Flow in HDFS, developing a Map Reduce Application-How Map Reduce Works.

**UNIT-III:**

**Frameworks:** Applications on Big Data Using Pig and Hive – Data processing operators in Pig – Hive services –HiveQL – Querying Data in Hive - fundamentals of HBase and Design of Zookeeper.

**UNIT-IV:**

**Introduction to Tableau:** Connecting Tableau to different sources, Understanding the data, Navigating Tableau, drawing basic graphs, understanding show me card, understanding different shelves, understanding marks card, Understanding Aggregation, Granularity, and level of detail, Creating Calculated Fields

**UNIT-V:**

**Data Visualization:** Prepare the data for Visualization, use tools like Tableau, Quick View and D3, Draw insights out of Visualization tool.

**TEXT BOOKS:**

1. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbach, Pearson
2. Mining of Massive Datasets, Anand Rajaraman and Jeffrey David Ullman, Cambridge University Press, 2012
3. Hadoop: The Definitive Guide, Tom White, 3<sup>rd</sup> Edition, O'Reilly Media, 2012

**REFERENCES:**

1. Making Sense of Data, Glenn J. Myatt, John Wiley & Sons, 2007
2. Big Data Glossary, Pete Warden, O'Reilly, 2011
3. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, 2<sup>nd</sup> Edition, Elsevier, 2008

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PC1DS402) DEEP LEARNING APPLICATIONS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To develop and apply major deep learning techniques, the problem settings, and their applications to solve real world problems
- To identify the deep learning algorithms which are more appropriate for various types of learning tasks in various domains
- To correlate and apply the comfortable of tools and techniques required in handling large amounts of datasets

**COURSE OUTCOMES:** After completion of the course, the student will be able to

**CO1:** Understand and apply the multi-layered perceptron for solving the stochastically complex problems

**CO 2:** Identify and evaluate the speed performance and dimension reduction using the SGD and auto encoder

**CO 3:** Apply and analyze various activation function with GPU in deep learning using Tensor flow and Keras

**CO 4:** Relate and build deep learning algorithms and solve real-world problems

**CO 5:** Analyze the case study of music generation system using deep learning

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 2                     | 2    | 2    | 2    | 2    | 2    | 1    | 3    | 2    | 2     | 1     | 2     | 2                               | 3     | 3     | 1     |
| CO-2 | 1                     | 1    | 3    | 3    | 1    | 1    | -    | -    | 3    | 3     | -     | 2     | 3                               | 3     | 3     | 1     |
| CO-3 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | -    | 3    | 3     | -     | 3     | 2                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 3    | 2    | 1    | 3    | 3    | 3     | 2     | 3     | 2                               | 2     | 3     | 2     |
| CO-5 | 1                     | 2    | 2    | 3    | 3    | 3    | -    | 3    | 3    | 3     | 2     | 3     | 3                               | 3     | 2     | 2     |

#### UNIT-I:

Diagrammatic representation Logistic Regression and Perceptron, Multi-Layered Perceptron (MLP) Notation, Training a single-neuron model, Training an MLP Chain Rule, training an MLP Memorization, back propagation, activation functions, vanishing gradient problem, bias variance trade-off, decision surfaces playground.

#### UNIT-II:

Deep Multi-layer perceptron's 1980s to 2010s, Dropout layers & Regularization, Rectified Linear Units (ReLU), Weight initialization, Batch Normalization, Optimizers Hill-descent analogy in 2D, Optimizers Hill descent in 3D and contours. SGD Recap, Batch SGD with momentum, Nesterov Accelerated Gradient (NAG)

**UNIT-III:**

Optimizers Ada Grad, Optimizers Ad delta and RMS Prop, Adam, Which algorithm to choose when, Gradient Checking and clipping, Softmax and Cross-entropy for multi-class classification, How to train a Deep MLP, Auto Encoders, Word2Vec CBOW, Word2Vec Skip-gram, Word2Vec Algorithmic Optimizations

**UNIT-IV:**

Tensor flow and Keras overview, GPU vs CPU for Deep Learning, Google Colaboratory, Install Tensor Flow, Online documentation, Softmax Classifier on MNIST dataset, MLP Initialization, Model 1 Sigmoid activation, Model 2ReLU activation, Model 3 Batch Normalization, Model 4 Dropout, MNIST classification in Keras, Hyper parameter tuning in Keras.

**UNIT-V:**

**Case Study:** Music Generation using Deep Learning:

Real-world problem-MUSIC, Music representation, Char-RNN with abc-notation Char-RNN model, Char-RNN with abc-notation Data preparation, Char-RNN with abc-notation Many to Many RNN,Time Distributed-Dense layer,Char-RNN with abc-notation State full RNN,Char-RNN with abc-notation Model architecture. Model training, Char-RNN with abc-notation music generation, Char-RNN with abc-notation with Tabla music, MIDI music generation, Survey blog.

**TEXT BOOKS:**

1. Deep Learning, Goodfellow, I., Bengio Y., Courville A., MIT Press, 2016
2. Deep Learning: A Practitioner's Approach, Josh Patterson, Adam Gibson, O'Reilly Media, 2017

**REFERENCES:**

1. Machine Learning with Tensor Flow, Nishant Shukla
2. Deep Learning with Keras, Antonio Gulli and Sujit Pal

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1IN303) ADHOC AND SENSOR NETWORKS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn about the issues and challenges of wireless Adhoc networks
- To understand the working of MAC and Routing Protocols for Adhoc and sensor networks
- To learn about the Transport Layer protocols and their QoS for Adhoc and sensor networks
- To understand various security issues in Adhoc and sensor networks and the corresponding solutions
- To understand the various energy management schemes in Adhoc and sensor networks

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand basic concepts and challenges related to Adhoc sensor networks

**CO-2:** Analyze architecture and issues in designing protocols for Adhoc and sensor networks

**CO-3:** Identify and analyze various quality services and security issues of Adhoc and sensor networks

**CO-4:** Relate the applications and energy management schemes for Ad hoc sensor networks

**CO-5:** Implement different protocols in real time applications

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 2    | 1    | 1    | 2    | 1    | 2    | 2    | 2     | 2     | 1     | 1                               | 2     | 1     | 1     |
| CO-2 | 2                     | 2    | 3    | 1    | 2    | 3    | 1    | 3    | 1    | 1     | 2     | 2     | 2                               | 1     | 3     | 1     |
| CO-3 | 2                     | 2    | 2    | 1    | 3    | 1    | 1    | 1    | 3    | 2     | 2     | 2     | 3                               | 1     | 2     | 2     |
| CO-4 | 2                     | 1    | 3    | 2    | 1    | 1    | 1    | 3    | 2    | 2     | 2     | 2     | 1                               | 1     | 2     | 2     |
| CO-5 | 2                     | 2    | 2    | 2    | 1    | 2    | 1    | 1    | 1    | 1     | 1     | 2     | 1                               | 1     | 3     | 2     |

#### UNIT-I:

**Introduction to AdHoc Wireless Networks:** Introduction, Issues in AdHoc Wireless Networks. Ad Hoc Wireless Internet. Summary. Problems. Bibliography.

**MAC Protocols for AdHoc Wireless Networks:** Introduction. Issues in Designing a MAC Protocol for Ad Hoc Wireless Networks. Design Goals of a MAC Protocol for Ad Hoc Wireless Networks. Classifications of MAC Protocols. Contention-Based Protocols. Contention-Based Protocols with Reservation Mechanisms. Contention-Based MAC Protocols with Scheduling Mechanisms. MAC Protocols That Use Directional Antennas. Other MAC Protocols

**UNIT-II:**

**Routing Protocols for AdHoc Wireless Networks:** Introduction. Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks. Classifications of Routing Protocols. Table-Driven Routing Protocols. On-Demand Routing Protocols. Hybrid Routing Protocols. Routing Protocols with Efficient Flooding Mechanisms. Hierarchical Routing Protocols. Power-Aware Routing Protocols.

**UNIT-III:**

**Multi cast routing in AdHoc Wireless Networks:** Introduction. Issues in Designing a Multicast Routing Protocol. Operation of Multicast Routing Protocols. An Architecture Reference Model for Multicast Routing Protocols. Classifications of Multicast Routing Protocols. Tree-Based Multicast Routing Protocols. Mesh-Based Multicast Routing Protocols. Summary of Tree-and Mesh-Based Protocols. Energy-Efficient Multicasting. Multicasting with Quality-of-Service Guarantees. Application-Dependent Multicast Routing.

**UNIT-IV:**

**Transport Layer and Security Protocols for AdHoc Wireless Networks:** Introduction. Issues in Designing a Transport Layer Protocol for Ad Hoc Wireless Networks. Design Goals of a Transport Layer Protocol for Ad Hoc Wireless Networks. Classification of Transport Layer Solutions. TCP Over Ad Hoc Wireless Networks. Other Transport Layer Protocols for Ad Hoc Wireless Networks. Security in Ad Hoc Wireless Networks. Network Security Requirements. Issues and Challenges in Security Provisioning. Network Security Attacks. Key Management. Secure Routing in Ad Hoc Wireless Networks

**UNIT-V:**

**Quality of Service and Energy Management in AdHoc Wireless Networks:** Classifications of QoS Solutions. MAC Layer Solutions. Network Layer Solutions. QoS Frameworks for Ad Hoc Wireless Networks, Classification of Energy Management Schemes. Battery Management Schemes. Transmission Power Management Schemes. System Power Management Schemes.

**TEXT BOOKS:**

1. AdHoc Wireless Networks – Architectures and Protocols, C. Siva Ram Murthy and B. S. Manoj, Pearson Education, 2006
2. Protocols and Architectures for Wireless Sensor Networks, Holger Karl, Andreas Willing, John Wiley & Sons, 2005

**REFERENCES:**

1. AdHoc Mobile Wireless Networks, Subir Kumar Sarkar, T. G. Basavaraju, C. Puttamadappa, Auerbach Publications, 2008
2. AdHoc and Sensor Networks: Theory and Applications, Carlos De Moraes Cordeiro, Dharma Prakash Agrawal, 2<sup>nd</sup> Edition, World Scientific, 2011
3. Fundamentals of Wireless Sensor Networks Theory and Practice, Waltenegus Dargie, Christian Poellabauer, John Wiley and Sons, 2010
4. Wireless AdHoc and Sensor Networks: Theory and Applications, Xiang-Yang Li, Cambridge University Press, 2008



# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1IT401) SOFT COMPUTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Basic knowledge of Machine Learning

#### COURSE OBJECTIVES:

- To familiarize with soft computing concepts, technologies and applications
- To introduce and use the idea of fuzzy logic and use of heuristics based on human experience
- To familiarize the fuzzy modeling
- To learn the concepts of genetic algorithm and its applications
- To acquire the knowledge of hybrid systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify the difference between conventional artificial intelligence to computational intelligence

**CO-2:** Understand fuzzy logic and reasoning to handle and solve engineering problems

**CO-3:** Apply the fuzzy logic and its applications

**CO-4:** Perform various operations of genetic algorithm

**CO-5:** Understand hybrid systems

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 2    | 1    | 1    | 2    | 1    | 2    | 2    | 2     | 2     | 1     | 2                               | 1     | 2     | 1     |
| CO-2 | 2                     | 2    | 3    | 2    | 2    | 3    | 1    | 3    | 1    | 2     | 2     | 2     | 3                               | 2     | 3     | 1     |
| CO-3 | 2                     | 2    | 2    | 3    | 3    | 1    | 2    | 2    | 3    | 2     | 2     | 2     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 3    | 2    | 2     | 2     | 2     | 2                               | 3     | 2     | 2     |
| CO-5 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 1    | 1    | 2     | 2     | 2     | 3                               | 2     | 3     | 2     |

#### UNIT-I:

**Introduction to Soft Computing:** Evolutionary Computing, "Soft" computing versus "Hard" computing, Soft Computing Constituents and Conventional Artificial Intelligence, Characteristics of Soft computing, Applications of Soft Computing Techniques.

#### UNIT-II:

**Fuzzy Systems:** Introduction to Fuzzy Logic, Classical Sets (Crisp Sets), Fuzzy Sets, Basic operations on Fuzzy Sets, Fuzzy Membership functions, Crisp Relations and Fuzzy Relations.

**UNIT-III:**

**Fuzzy Rule Base and Approximate Reasoning:** Fuzzy Inference Systems (FIS)-Mamdani and Takagi and Sugeno architectures, Fuzzification and Defuzzification methods, Fuzzy Decision Making.

**UNIT-IV:**

**Genetic Algorithm:** Genetic Algorithm vs. Traditional optimization methods, Basic Terminologies in Genetic Algorithm, Basic Operators in Genetic Algorithm, Encoding, Selection, Crossover and Mutation, Advantages and Limitations of Genetic Algorithm, Applications of Genetic Algorithm.

**UNIT-V:**

**Hybrid Soft Computing Techniques:** Integration of Neural Networks, Fuzzy Logic and Genetic Algorithms, Neuro-Fuzzy Hybrid Systems, Genetic Neuro-Hybrid Systems, Genetic Fuzzy Hybrid and Fuzzy Genetic Hybrid Systems.

**TEXT BOOKS:**

1. Fuzzy Logic with Engineering Applications, Timothy J. Ross, 3<sup>rd</sup> Edition, John Wiley & Sons, 2010
2. Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis and Applications, S. Rajasekaran, G. A. Vijayalakshmi Pai, Prentice Hall of India, 2007
3. Neuro-Fuzzy and Soft Computing, J. S. R. Jang, C. T. Sun and E. Mizutani, Prentice Hall of India, 2009

**REFERENCES:**

1. Fuzzy Sets and Fuzzy Logic Theory and Applications, George J. Klir and Bo Yuan, Prentice Hall, New Jersey, 1995
2. Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence, Jyh-Shing Roger Jang, Chuen-Tsai Sun, Eiji Mizutani, 2<sup>nd</sup> Edition, Prentice Hall of India, 2002
3. Genetic Algorithms- In Search, Optimization and Machine Learning, D. E. Goldberg, 2<sup>nd</sup> Edition, Addison Wesley, 2007
4. An Introduction to Genetic Algorithms, Melanie Mitchell, MIT Press, 2000

**ONLINE RESOURCES:**

1. <https://link.springer.com/journal/500>
2. [https://www.youtube.com/watch?v=K9gjuXjJeEM&list=PLJ5C\\_6qdAvBFqAYS0P9INAogIMklG8E-9](https://www.youtube.com/watch?v=K9gjuXjJeEM&list=PLJ5C_6qdAvBFqAYS0P9INAogIMklG8E-9)

**B.Tech. VII Semester**

**(22PE1DS402) DISTRIBUTED COMPUTING**

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE OBJECTIVES:**

- To introduce concepts related to distributed computing systems
- To focus on performance and flexibility issues related to systems design decisions
- To expose students to current literature in distributed systems
- To prepare students for a real-world distributed application design

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand models of distributed computing

**CO-2:** Analyze algorithms for coordination, communication and synchronization in distributed systems

**CO-3:** Analyze distributed shared memory models

**CO-4:** Design and implement distributed file systems

**CO-5:** Design distributed algorithms for handling deadlocks

**COURSE ARTICULATION MATRIX:**

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 2                     | 2    | 1    | 1    | -    | 1    | 1    | 1    | -    | 1     | -     | 1     | 1                               | -     | 1     | 1     |
| CO-2 | 2                     | 2    | 1    | 1    | 1    | 1    | 1    | 1    | -    | -     | -     | -     | 1                               | 1     | 1     | 1     |
| CO-3 | 3                     | 3    | 2    | 2    | 2    | 1    | 1    | 1    | -    | -     | -     | -     | 1                               | 1     | 1     | 2     |
| CO-4 | 1                     | 1    | 3    | 3    | 2    | 1    | 1    | 1    | -    | -     | 1     | 1     | 1                               | -     | 1     | 2     |
| CO-5 | 1                     | -    | 1    | 1    | 2    | 2    | 1    | 1    | -    | 1     | -     | 2     | 1                               | -     | 1     | 2     |

**UNIT-I:**

**Distributed Computing Introduction:** Definition, Relation to parallel systems, synchronous vs asynchronous execution, design issues and challenges .A Model of Distributed Computations: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Logical Time: Logical clocks, scalar time, vector time, Matrix time, virtual time, Physical clock synchronization – NTP.

**UNIT-II:**

**Global State and Snapshot Recording Algorithms:** System model, Snapshot algorithms for FIFO channels, Variations of Chandy-Lamport algorithm, Snapshot algorithms for non-FIFO channels, Snapshots in a causal delivery system, Monitoring global state, Necessary and sufficient conditions for consistent global snapshots, Finding consistent global snapshots in a distributed computation.

**Message Ordering and Group Communication:** Message ordering paradigms, Group communication, Causal order (CO), Total order, Multicast, Propagation trees for multicast, Application-level multicast algorithms, Fault-tolerant group communication, Multicast algorithms at the network layer.

### UNIT-III:

**Termination Detection:** System model of a distributed computation, Termination detection using distributed snapshots, weight throwing and spanning-tree-based algorithms, Message-optimal termination detection, and Termination detection in a general distributed computing model, Termination detection in the atomic computation model, Termination detection in a faulty distributed system.

**Distributed Mutual Exclusion Algorithms:** Lamport's algorithm, Ricart–Agrawala algorithm, Singhal's dynamic information-structure algorithm, Lodha and Kshemkalyani's fair mutual exclusion algorithm, Quorum-based mutual exclusion algorithms, Maekawa's algorithm, Agarwal–El Abbadi quorum-based algorithm, Token-based algorithms, Suzuki–Kasami's broadcast algorithm, Raymond's tree-based algorithm.

### UNIT-IV:

**Deadlock Detection in Distributed Systems:** System model, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single resource model, Chandy–Misra–Haas algorithm for the AND model, Chandy–Misra–Haas algorithm for the OR model, Kshemkalyani–Singhal algorithm for the P-out-of-Q model .

**Distributed Shared Memory:** Abstraction and advantages, Memory consistency models, Shared memory mutual exclusion, Wait-freedom, Register hierarchy and wait-free simulations, Wait-free atomic snapshots of shared objects

### UNIT-V:

**Check Pointing and Rollback Recovery:** Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Koo–Toueg coordinated check pointing algorithm, Juang–Venkatesan algorithm for asynchronous check pointing and recovery, Manivannan–Singhal quasi-synchronous check pointing algorithm, Peterson–Kearns algorithm based on vector time, Helary–Mostefaoui–Netzer–Raynal communication-induced protocol.

### TEXT BOOKS:

1. Distributed Computing, Ajay D. Kshemakalyani, Mukesh Singhal, Cambridge University Press, 2008
2. Distributed Systems - Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, Prentice Hall of India, 2004

### REFERENCES:

1. Distributed Computing: Fundamentals, Simulations, and Advanced Topics, Hagit Attiya, Jennifer Welch, Wiley
2. Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen, Prentice Hall of India

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1IT402) SOFTWARE PROJECT MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Software Engineering

#### COURSE OBJECTIVES:

- To identify and discuss the conventional and contemporary software project management principles
- To assess and plan project schedule and assign resources
- To apply an appropriate project development methodology among various alternating processes
- To identify project risks, understand the responsibilities, monitor and track project deadlines and the capability to work in a team environment

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Describe the conventional s/w management and explain how to improve s/w economics

**CO-2:** Identify and discuss the key phases of project management and the key skills associated with each

**CO-3:** Relate an appropriate project management approach through an evaluation of context and project scope and knowledge of agile and traditional and Global project management approaches, risk and quality management

**CO-4:** Apply the knowledge of the key project management skills, such as product and work break-down structure, schedule; governance, progress reporting and People Focused Process Models

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 2    | 1    | 1    | 2    | 1    | 2    | -    | -     | -     | 1     | 2                               | 1     | 2     | 1     |
| CO-2 | 2                     | 2    | 3    | 2    | 2    | 3    | 1    | 3    | -    | -     | -     | 2     | 3                               | 2     | 3     | 1     |
| CO-3 | 2                     | 2    | 2    | 3    | 3    | 1    | 2    | 2    | -    | -     | -     | 2     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 3    | -    | -     | -     | 2     | 2                               | 3     | 2     | 2     |

#### UNIT-I:

**Conventional Software Management:** The waterfall model, Conventional software Management performance. Overview of Project Planning – Stepwise Project Planning.

**Improving Software Economics:** Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections.

**UNIT-II:**

**The Old Way and the New Way:** The principles of conventional software Engineering, Principles of modern software management, Transitioning to an iterative process.

**Life Cycle Phases:** Engineering and Production stages, Inception, Elaboration, Construction, Transition phases.

**Artifacts of the process:** The artifact sets, Management artifacts, Engineering artifacts, Pragmatic artifacts.

**Workflows of the Process:** Software process workflows, Iteration workflows.

**UNIT-III:**

**Checkpoints of the Process:** Major milestones, Minor Milestones, Periodic status assessments.

**Iterative Process Planning:** Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

**UNIT-IV:**

**Project Organizations and Responsibilities:** Line-of-Business Organizations, Understanding Behavior - Organizational Behavior, Project Organizations, Evolution of Organizations.

**Process Automation:** Automation building blocks, The Project Environment.

**UNIT-V:**

**Project Control and Process Instrumentation:** The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation

**Future Software Project Management:** Modern Project Profiles, Next generation Software economics, modern process transitions.

Emerging Trends in Project Management. Globalization Issues in Project management, People Focused Process Models.

**TEXT BOOKS:**

1. Software Project Management, Walker Royce, Pearson Education, 2005
2. Managing and global Software Projects, Ramesh Gopalaswamy, Tata McGraw-Hill, 2003

**REFERENCES:**

1. Software Project Management, Bob Hughes and Mike Cotterell, Tata McGraw-Hill Edition
2. Software Project Management, Joel Henry, Pearson Education
3. Software Project Management in Practice, Pankaj Jalote, Pearson Education, 2005

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1CS401) ADVANCED DATABASES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introducing distributed database management system and its design issues
- To exploring several algorithms for processing queries and be able to use them
- To describe the methods to translate complex conceptual data models into logical and physical database designs
- To demonstrating query optimization and its algorithms
- To enumerating the concepts behind distributed transaction processing

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand data processing and architecture of distributed database

**CO-2:** Apply partitioning and fragmentation techniques to the data distributed across the network

**CO-3:** Design and optimize query processing strategies on distributed fragments

**CO-4:** Explain transaction processing and concurrency control in distributed databases

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 3                     | 1    | 1    | -    | -    | 1    | 2    | -    | -    | -     | 1     | 1     | 2                               | 2     | 2     | 1     |
| CO-2 | 2                     | 2    | 3    | 3    | 1    | 1    | -    | -    | 1    | -     | 1     | 1     | 3                               | 2     | 2     | -     |
| CO-3 | 3                     | 3    | 3    | 3    | 3    | 2    | 2    | -    | 2    | -     | 2     | 3     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 2    | 2    | 1    | 3    | 2    | -    | 2    | -     | -     | 2     | 3                               | 3     | 2     | 3     |

#### UNIT-I:

**Introduction:** Distributed Data Processing, Distributed Database System, Promises of DDBSs, Problem areas.

#### UNIT-II:

**Distributed DBMS Architecture:** Architectural Models for Distributed DBMS, DDMBS Architecture.

**Distributed Database Design:** Alternative Design Strategies, Distribution Design issues, Fragmentation, Allocation.

#### UNIT-III:

**Query Processing and Decomposition:** Query Processing Objectives, Characterization of query processors, layers of query processing, query decomposition, Localization of distributed data.

**UNIT-IV:**

**Distributed Query Optimization:** Query optimization, centralized query optimization, Distributed query optimization algorithms.

**UNIT-V:**

**Transaction Management:** Definition, properties of transaction, types of transactions. Distributed concurrency control: Serializability, concurrency control Mechanisms & Algorithms, Time stamped & Optimistic concurrency control Algorithms, Deadlock Management.

**TEXT BOOKS:**

1. Principles of Distributed Database Systems, M. Tamer OZSU and Patuck Valduriez, Pearson, 2001
2. Distributed Databases, Stefano Ceri and Willipse Pelagatti, McGraw-Hill

**REFERENCES:**

1. Database System Concepts, Henry F. Korth, A Silberchatz and Sudershan, McGraw-Hill
2. Database Management Systems, Raghuramakrishnan and Johhanes Gehrke, McGraw-Hill
3. Readings in Database Systems, M. Stonebraker, 2<sup>nd</sup> Edition, Morgan Kauffman, 1993



# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PC2DS303) BIG DATA COMPUTING TECHNOLOGIES LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

#### COURSE OBJECTIVES:

- To explore and understand various HDFS commands in Hadoop environment
- To understand the concept of Hadoop framework and its components
- To explore various Big Data tools for data processing
- To analyse data using visualization tools
- To apply data visualization tools such as Tableau, Quick View and D3

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explore various HDFS commands on Hadoop file system

**CO-2:** Illustrate map reduce programming framework

**CO-3:** Apply large data sets in a distributed environment

**CO-4:** Demonstrate different Big Data tools such as HIVE, PIG, and SGOOP

**CO-5:** Extract large data sets to perform visualization and data analysis tasks

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 3                     | -    | 2    | -    | 3    | -    | -    | -    | 2    | -     | 1     | 2     | 3                               | 2     | -     | 1     |
| CO-2 | 2                     | -    | 2    | -    | 3    | -    | -    | -    | 2    | -     | 3     | 2     | 3                               | 2     | -     | 1     |
| CO-3 | 3                     | -    | 2    | -    | 3    | -    | -    | -    | 2    | -     | 2     | 2     | 3                               | 2     | -     | 2     |
| CO-4 | 2                     | -    | 2    | -    | 3    | -    | -    | -    | 2    | -     | 1     | 2     | 3                               | 2     | -     | 2     |
| CO-5 | 2                     | -    | 2    | -    | 2    | -    | -    | -    | 2    | -     | 2     | 2     | 3                               | 2     | -     | 2     |

#### LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

##### 1. HDFS (Storage)

###### A. Hadoop Storage File system

Your first objective is to create a directory structure in HDFS using HDFS commands. Create the local files using Linux commands and move the files to HDFS directory. and vice versa.

I. Write a command to create the directory structure in HDFS.

II. Write a Command to move file from local Unix/Linux machine to HDFS.

###### B. Viewing Data Contents, Files and Directory

Try to perform these simple steps:

a) Write HDFS command to see the contents of files in HDFS.

b) Write HDFS command to see contents of files which are present in HDFS.

###### C. Getting Files data from the HDFS to Local Disk

I. Write an HDFS command to copy the file from HDFS to the local file system.

To process any data first move the data in HDFS. All files storage in HDFS can be.

accessed using HDFS commands.

**Lab Instructions:**

- i. Your objective is to use HDFS commands to move data to HDFS for processing data.

## **2. Map Reduce Programming (Processing data).**

Hadoop Map-Reduce framework is developed using Java, but the framework allows you to write programs in other languages as well.

### **A. Word Count**

The word count problem is the most famous using map reduce program. Same thing We can do it with java but takes lot of time with huge files, in MR it will process less time.

even with huge and distributed files. The objective is to count the frequency of words of a large text.

**Lab Instructions:**

- i. Develop MapReduce example program in a MapReduce environment to find out the number of occurrences of each word in a text file

### **B. Most Frequent Words Count**

Use the output from the previous program to list the most frequent words with their counts.

**Lab Instructions:**

- i. Use the same strategy of breaking the programs in three parts. Copy the files from the previous exercise and use them as a starting point.
- ii. Load the data from the output by using third filter to load the files.

## **3. Data Processing Tool – Hive (NOSQL query-based language)**

Hive command line tool allows you to submit jobs via bash scripts.

Identifying properties of a data set:

We have a table 'user data' that contains the following fields:

data date: string

user\_id: string

properties: string

The properties field is formatted as a series of attribute=value pairs.

Ex: Age=21; state=CA; gender=M;

**Lab Instructions:**

- i. Create the table in HIVE using hive nosql based query.
- ii. Fill the table with sample data by using some sample databases.
- iii. Write a program that produces a list of properties with minimum value(min\_value), largest value(max\_value) and number of unique values. Before you start, execute the prepare step to load the data into HDFS.
- iv. Generate a count per state.

Now that extracted the properties, calculate the number of records per state.

**Lab Instructions:**

- i. Write a program that lists the states and their count from the data input.

## **4. Data Processing Tool – Pig (Latin based scripting lang)**

Pig command line tool like the Hive allows you to submit jobs via bash scripts.

### **A. Simple Logs**

We have a set of log files and need to create a job that runs every hour and perform some calculations.

The log files are delimited by a 'tab' character and have the following fields:

a) site

b) hour\_of\_day

- c) page\_views
- d) data\_date

The log files are located on the prepare folder. Load them in HDFS at data/ pig/ simple logs folder and use them as the input.

Important: In order to load tab delimited files use pigStorage ('\u0001').

#### **Lab Instructions:**

Create a program to:

- i. Calculate the total views per hour per day.
- ii. Calculate the total views per day.
- iii. Calculate the total counts of each hour across all days.
- iv. We can write word count script by passing text file as input.

### **5. SQOOP**

It is used to import and export data from SQL to HDFS and Vice versa.

Before processing any data first need to import data into HDFS. As of now seen importing files into HDFS but using sqoop tool we can import SQL table data into hive table.

- I. Create table in HIVE using hive query language.
- II. Import the sql table data into hive using sqoop too.
- III. Export hive table data into local machine and into SQL.

### **6. Exploring ARIMA model.**

#### **TEXT BOOKS:**

- 1. Data Analytics using R, Seema Acharya, McGraw-Hill, 2019
- 2. R for Data Science, Import, Tidy, Transform, Visualize, and Model Data, Hadley Wickham, Garrett Golemund, 2016
- 3. Hadoop: The Definitive Guide, Tom White, 3<sup>rd</sup> Edition, O'Reilly Media, 2012

#### **REFERENCES:**

- 1. Data Analytics with R, Bharti Motwani, 2019
- 2. Data Science-Theory, Analysis and Applications, Qurban A. Memon, Shakeel Ahmed Khoja, 2019
- 3. Big Data Glossary, Pete Warden, O'Reilly, 2011

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PC2DS402) DEEP LEARNING APPLICATIONS LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the fundamental principles of autonomous vehicles
- To develop skills to both design and critique visualizations
- To obtain a comprehensive knowledge of various techniques for deep learning
- To analyse the various case studies of data analysis

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the fundamentals of autonomous vehicles like self-driving cars

**CO-2:** Learn the basics of deep learning techniques and how they are applied to HAR

**CO-3:** Equip students with the knowledge and skills necessary to understand and solve the Quora Question Pair Similarity problem

**CO-4:** Grasp the foundational concepts of music generation using deep learning techniques

**CO-5:** Learn to design the models of the problem statements using RNN & CNN

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 2                     | 2    | 2    | 2    | 2    | 2    | 1    | 3    | 2    | 2     | 1     | 2     | 2                               | 3     | 3     | 1     |
| CO-2 | 1                     | 1    | 3    | 3    | 1    | 1    | -    | -    | 3    | 3     | -     | 2     | 3                               | 3     | 3     | 1     |
| CO-3 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | -    | 3    | 3     | -     | 3     | 2                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 3    | 2    | 1    | 3    | 3    | 3     | 2     | 3     | 2                               | 2     | 3     | 2     |
| CO-5 | 1                     | 2    | 2    | 3    | 3    | 3    | -    | 3    | 3    | 3     | 2     | 3     | 3                               | 3     | 2     | 2     |

#### LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

##### WEEK 1-4: Self driving car in deep learning

- Problem definition and dataset
- Data understandings analysis files and folders
- Dash-com, images and steering analysis
- EDA steering angles
- Mean baseline model
- Deep learning model, Deep learning for regression
- Batch load the dataset
- NVIDIA's end to end CNN MODEL
- Train the model
- Test and visualize the output and its extensions

##### WEEK 5-8: Human activity recognition

- Problem definition
- Dataset understanding

- c) Data cleaning and pre-processing
- d) EDA univariate analysis
- e) Classical ML models
- f) Deep learning models

#### **WEEK 9-12: Quora question pair similarity problem**

- a) Business real world problem definition
- b) Business objectives and constraints
- c) Mapping to an ML problem data overview
- d) Mapping to an ML problem and performance measures
- e) Mapping to an ML problem train-test split
- f) EDA basic statistics, feature extraction
- g) EDA text processing, advanced feature extraction
- h) EDA feature analysis, data visualization
- i) EDA IT-IDF weighting word2vec factorization
- j) ML models loading, random model
- k) ML models logistics regressions and linear SVM

#### **WEEK 13-14: Music generation using deep learning**

- a) Real-world problem and music representation
- b) Char-RNN with abc –notation char CNN model
- c) Char-RNN with abc –notation data preparation
- d) Char-RNN with abc –notation many to many RNN time distribution
- e) Char-RNN with abc –notation state full RNN,model architecture
- f) Char-RNN with abc –notation music generation, generate tabla music.
- g) MIDI music generation.

#### **WEEK 15: LAB INTERNAL**

##### **TEXT BOOKS:**

1. Computational and Inferential Thinking: The Foundations of Data Science, Adi Adhikari and John De Nero, 1<sup>st</sup> Edition, 2019
2. Doing Data Science, Straight Talk from The Frontline, Cathy O'Neil and Rachel Schutt, O'Reilly, 2014
3. Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures, Claus Wilke, 1<sup>st</sup> Edition, O'Reilly Media, 2019

##### **REFERENCES:**

1. Mining of Massive Datasets, v2.1, Jure Leskovek, Anand Rajaraman and Jeffrey Ullman, Cambridge University Press, 2014
2. Practical Statistics for Data Scientists: 50 Essential Concepts, Bruce, Peter, and Andrew Bruce, O'Reilly Media, 2017
3. Data Analysis and Visualization Using Python: Analyze Data to Create Visualizations for BI Systems, Ossama Embarak, Press, 201

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1IT403) ADVERSARIAL MACHINE LEARNING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PREREQUISITES:** Basic knowledge of Machine Learning

#### COURSE OBJECTIVES:

- To understand the basic notation and introduce the framework for secure learning and the various phases involved in the learning process
- To study causative attacks on machine learning algorithms and their impact on learning
- To examine a case study on causative attacks against the SpamBayes spam filter and understand the defense mechanisms used
- To explore different types of attacks on specific machine learning models such as hypersphere learners and PCA detectors
- To delve into exploratory attacks on machine learning, focusing on privacy-preserving mechanisms for SVM learning

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basic notation and how security analysis fits into the learning process

**CO-2:** Apply the gained knowledge to defend machine learning models against causative attacks

**CO-3:** Create a practical experience in corrupting PCA subspaces

**CO-4:** Evaluate the SpamBayes spam filter and its susceptibility to causative attacks

**CO-5:** Analyse privacy-preserving mechanisms for SVM learning in protecting sensitive data

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 2    | 1    | 1    | 2    | 1    | 2    | -    | -     | -     | 1     | 2                               | 1     | 2     | 1     |
| CO-2 | 2                     | 2    | 3    | 2    | 2    | 3    | 1    | 3    | -    | -     | -     | 2     | 3                               | 2     | 3     | 1     |
| CO-3 | 2                     | 2    | 2    | 3    | 3    | 1    | 2    | 2    | -    | -     | -     | 2     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 3    | -    | -     | -     | 2     | 2                               | 3     | 2     | 2     |
| CO-5 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 1    | -    | -     | -     | 2     | 3                               | 2     | 3     | 2     |

#### UNIT-I:

##### Introduction

**Background and Notation:** Basic Notation, Statistical Machine Learning.

**A Framework for Secure Learning:** Analysing the Phases of Learning, Security Analysis, Framework, Exploratory Attacks, Causative Attacks.

**UNIT-II:****Causative Attacks on Machine Learning**

**Attacking a Hypersphere Learner:** Causative Attacks on Hypersphere Detectors, Hypersphere Attack Description, Optimal Unconstrained Attacks, Imposing Time Constraints on the Attack, Constrained Attacks.

**UNIT-III:**

**Availability Attack Case Study: Spam Bayes:** The SpamBayes Spam Filter, Threat Model for SpamBayes, Causative Attacks against SpamBayes' Learner, the Reject on Negative Impact (RONI) Defense, Experiments with SpamBayes.

**UNIT-IV:**

**Integrity Attack Case Study: PCA Detector:** PCA Method for Detecting Traffic Anomalies, Corrupting the PCA Subspace, Corruption-Resilient Detectors, Empirical Evaluation.

**UNIT-V:****Exploratory Attacks on Machine Learning**

**Privacy-Preserving Mechanisms for SVM Learning:** Privacy Breach Case Studies, Problem Setting: Privacy-Preserving Learning, Support Vector Machines: A Brief Primer.

**TEXT BOOK:**

1. Adversarial Machine Learning, Anthony D. Joseph, Blaine Nelson, Benjamin I. P. Rubinstein, J. D. Tygar, Cambridge University Press, 2019

**REFERENCES:**

1. Adversarial Machine Learning, Vol. 12, Vorobeychik, Yevgeniy, Murat Kantarcioglu, Ronald Brachman, Peter Stone, and Francesca Rossi, San Rafael, Morgan & Claypool Publishers, 2018
2. Generative Deep Learning: Teaching Machines to Paint, Foster David, Write, Compose, and Play (Japanese Version) O'Reilly Media, 2019, 139-140

**ONLINE RESOURCES:**

1. <https://www.udacity.com/blog/2023/03/the-role-of-ai-and-ml-in-cybersecurity.html>
2. <https://www.youtube.com/watch?v=3-qazNQS2JU>

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1CS403) COMPUTER VISION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce the basic concepts of computer vision
- To explore the low-level feature extraction methods
- To understand the deep learning models for computer vision
- To introduce 3D computer vision and its applications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Describe the basic computer vision concepts

**CO-2:** Extract the feature extraction methods

**CO-3:** Understand the deep learning models for computer vision

**CO-4:** Explore and identify 3D computer vision applications

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 1    | 1    | 1    | 2    | -    | -    | -    | -     | -     | -     | 1                               | 1     | 1     | 1     |
| CO-2 | 2                     | 3    | 3    | 3    | 3    | 3    | -    | -    | -    | -     | -     | -     | 3                               | 3     | 3     | 3     |
| CO-3 | 3                     | 3    | 3    | 3    | 3    | 3    | 1    | -    | -    | -     | -     | -     | 3                               | 3     | 3     | 3     |
| CO-4 | 1                     | 1    | 1    | 1    | 1    | 2    | 1    | -    | -    | -     | -     | -     | 1                               | 2     | 1     | 1     |

#### UNIT-I:

**Digital Image Fundamentals:** Elements of visual perception, light and the electromagnetic spectrum, Image sensing, and Acquisition, Sampling and Quantization, Basic relationships between pixels, Gray level transformations, Histogram processing, Spatial filtering: smoothing and sharpening.

#### UNIT-II:

**Visual Feature Extraction:** Low-level Feature Extraction: Color histograms, First order edge detection, Second order edge detection, Scale Invariant Feature Transform (SIFT), Texture features, Mathematical morphology.

#### UNIT-III:

**Recognition:** Image Recognition, Image classification, Object detection, semantic segmentation, Video understanding, vision, and language.

#### UNIT-IV:

**Applications:** Object Detection, Surveillance application, Face recognition, Activity recognition, Segmentation in autonomous driving, Robot vision



**UNIT-V:**

**3D Computer vision:** Stereo-pair images and depth perception, 3D vision systems, 3D vision applications, Case examples: 3D systems for vision impaired persons, face and body modeling, clinical and veterinary applications, movie restoration.

**TEXT BOOKS:**

1. Digital Image Processing, Gonzalez, R. C, Pearson Education, 2009
2. Feature Extraction and Image Processing for Computer Vision, Nixon M. & Aguado A., Academic Press, 2019
3. Deep Learning in Computer Vision: Principles and Applications, Hassaballah M. & Awad A. I. (Ed.), CRC Press, 2020

**REFERENCES:**

1. An Introduction To 3D Computer Vision Techniques and Algorithms, Cyganek B., & Siebert J. P., John Wiley & Sons, 2011
2. Computer Vision: Algorithms and Applications, Szeliski R., Springer Nature, 2022

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1AM405) COGNITIVE ENGINEERING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To identify the basics of cognitive science engineering with focus on mind organization, brains
- To analyze human intelligence and artificial intelligence
- To understand basics of neuroscience and linguistics
- To analyze various levels of cognitive functioning and its applications
- To relate the concepts of cognitive field in robotics and real time applications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Introduce role of computers in cognitive science and neuroscience applications

**CO-2:** Illustrate various information processing models and cognitive architecture

**CO-3:** Apply various computation tools for cognitive functioning

**CO-4:** Analyze cognitive science applications in image processing linguistic applications

**CO-5:** Demonstrate the use of cognitive models and patterns in distributed environment, and real time applications

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 2    | 1    | 1    | 2    | 1    | 2    | -    | -     | -     | 1     | 2                               | 1     | 2     | 1     |
| CO-2 | 2                     | 2    | 3    | 2    | 2    | 3    | 1    | 3    | -    | -     | -     | 2     | 3                               | 2     | 3     | 1     |
| CO-3 | 2                     | 2    | 2    | 3    | 3    | 1    | 2    | 2    | -    | -     | -     | 2     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 3    | -    | -     | -     | 2     | 2                               | 3     | 2     | 2     |
| CO-5 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 1    | -    | -     | -     | 2     | 3                               | 2     | 3     | 2     |

#### UNIT-I:

**Introduction to Cognitive Science:** The Cognitive view –Some Fundamental Concepts – Computers in Cognitive Science – Applied Cognitive Science – The Interdisciplinary Nature of Cognitive Science, The Architecture of the Mind. Cognitive Processes, Computational Neuroscience - The Organization of the mind.

#### UNIT-II:

**Higher-Level Cognition and Challenges:** Dynamical systems and situated cognition-Challenges – Emotions and Consciousness – Computation of Cognitive Functioning in machines: Robotics, Human-Robotics Interaction.

#### UNIT-III:

**Cognitive Science for Vision/Image Processing:** Perception and sensing: visual cognition, cognitive mechanisms of vision, Feature Extraction from Images,

Information Processing in Perception and Visual Behavior, Cognitive function measurement tools and software.

#### **UNIT-IV:**

**Language Acquisition and Semantics:** Milestones in acquisition, Theoretical perspectives, Semantics and Cognitive Science, Meaning and entailment, Sense, Problems in possible worlds Semantics, Cognitive and computational models of semantic Processing.

#### **UNIT-V:**

**Distributed Cognitive Learning Using fuzzy Cognitive Maps:** Axelrod's cognitive maps, Kosko's model, adaptive FCM, Pal and Konar's FCM model-examples.

#### **TEXT BOOKS:**

1. Cognitive Science: An Introduction, Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, 2<sup>nd</sup> Edition, 1995
2. Cognitive Science: An Introduction to the Science of the Mind, José Luis Bermúdez, Cambridge University Press, 2010
3. Cognitive Engineering: A Distributed Approach to Machine Intelligence (Advanced Information and Knowledge Processing), Amit Konar, Springer, 2005

#### **REFERENCES:**

1. Computational Vision: Information Processing in Perception and Visual Behavior, Hanspeter A. Mallot, Translated by John S. Allen
2. How the Mind Works, Steven Pinker, 2009
3. Cognitive Science: An Interdisciplinary Approach, Carolyn Panzer Sobel and Paul Li, 2013
4. Mind: Introduction to Cognitive Science, Paul Thagard, 2<sup>nd</sup> Edition, MIT Press, 2004  
Wilson Robert A. & Keil Frank C. (Eds.)
5. The MIT Encyclopedia of the Cognitive Sciences (MITECS), MIT Press, 2001

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1CS404) AUGMENTED REALITY AND VIRTUAL REALITY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Programming and Data Structures

#### COURSE OBJECTIVES:

- To understand the basic concepts of Virtual and Augmented Reality
- To identify the differences in AR/VR concepts and technologies
- To understand the fundamental concepts relating to Virtual Reality such as presence, immersion, and engagement
- To evaluate usability of AR/VR applications and critique their use of AR/VR capabilities
- To design and prototype effective AR/VR applications using UNITY platform for business, industry, non-profit and government organizations

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Introduce role of augmented and virtual reality in interactive application development

**CO-2:** Apply Geometric Modeling Techniques

**CO-3:** Implement virtual environment with respect to user experience

**CO-4:** Design Virtual/Augmented Reality Applications

**CO-5:** Acquire fundamental and hands-on skills needed to develop AR/VR applications

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 1                     | 1    | 1    | 1    | 1    | 1    | 2    | 2    | 1    | -     | 2     | 2     | 3                               | 3     | 3     | 3     |
| CO-2 | 1                     | 1    | 1    | 3    | 1    | 2    | 2    | 1    | 2    | 1     | 1     | 2     | 3                               | 3     | 3     | 3     |
| CO-3 | 1                     | 1    | 2    | -    | 3    | 1    | 1    | 1    | 1    | 2     | 1     | 2     | 3                               | 3     | 3     | 3     |
| CO-4 | -                     | 2    | -    | -    | -    | 2    | -    | 2    | 1    | 1     | 3     | 1     | 3                               | 3     | 3     | 3     |
| CO-5 | -                     | -    | 2    | -    | 3    | -    | -    | -    | -    | -     | -     | -     | 3                               | 3     | 3     | 3     |

#### UNIT-I:

**Introduction to Augmented Reality (AR):** Definition and Scope A Brief History of Augmented Reality Displays (Multimodal Displays, Spatial Display Model, and Visual Displays) Strong vs Weak AR Applications of AR Challenges in AR

#### UNIT-II:

**Introduction to Virtual Reality (VR):** Definition and Scope Types of VR Characteristics of VR Basic VR environments Limitations of VR environments Immersion Vs Presence Key hardware requirements for VR

### **UNIT-III:**

**Interaction Design for AR/VR Environments:** Interaction design process Identifying user needs AR/VR design considerations Typical AR/VR Interface Metaphors Affordances in AR/VR Human Information Processing Design for Perception and cognition User experience (UX) guidelines for AR/VR UX challenges for AR/VR Prototyping for AR/VR Evaluation of the developed AR/VR prototype

### **UNIT-IV:**

**Unity Overview:** Windows, Interface, Navigation, Terminology, Game Objects, Hierarchy, Parenting Objects Asset Store, Importing Plug-ins Creating a Terrain, Materials, Colors, Transparency Introduction to MonoBehaviour: Awake, Start, Update

### **UNIT-V:**

#### **Introduction to Vuforia and Physics in UNITY**

**Vuforia Overview:** Interface, Navigation, Terminology, Image Targeting, Custom Images Overview of Physics in Unity

**Introduction to Scripting:** Terminology, Creating Objects, Accessing Components, Debugging, Lists, Loops

### **TEXT BOOKS:**

1. Understanding virtual reality, Sherman W. R., & Craig A. B., Morgan Kauffman, 2003
2. Augmented Reality: Principles and Practice, Schmalstieg D., & Hollerer T., Addison-Wesley Professional, 2016

### **REFERENCES:**

1. A survey of augmented reality. Presence: Teleoperators & Virtual Environments, Azuma R. T., 1997, 6(4), 355–385
2. Recent Advances in Augmented Reality, IEEE Computer Graphics and Applications, Azuma R., Baillot Y., Behringer R., Feiner S., Julier S., & MacIntyre B., 2001, 21(6), 34-47
3. A Cost-Effective Interactive 3D Virtual Reality System Applied To Military Live Firing Training, Virtual Reality, Bhagat K. K., Liou, W. K., & Chang, C. Y, 2016, 20(2), 127-140. doi:10.1007/s10055-016-0284-x
4. Augmented Reality Technologies, Systems And Applications, Multimedia Tools and Applications, Carmigniani J., Furht B., Anisetti M., Ceravolo P., Damiani E., & Ivkovic M., 2011, 51(1), 341-377
5. Human Augmentation: Past, Present and Future. International Journal Of Human Computer Studies, Raisamo R., Rakkolainen I., Majaranta P., Salminen K., Rantala J., & Farooq, 2019, 131, 131-143. doi: <https://doi.org/10.1016/j.ijhcs.2019.05.008>

### **ONLINE RESOURCES:**

1. Virtual Reality Engineering, Manivannan M., 2018, IIT Madras, <https://nptel.ac.in/courses/121106013>
2. Industry 4.0: Augmented Reality and Virtual Reality, Misra S., 2019, IIT Kharagpur, <https://www.youtube.com/watch?v=zLMgdYI82IE>
3. Augmented Reality - Fundamentals and Development, NPTEL Special Lecture Series, Dube A., 2020, <https://www.youtube.com/watch?v=MGuSTAqIZ9Q>
4. <http://cambum.net/course-2.htm>

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PC1CY401) BLOCKCHAIN TECHNOLOGY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To provide an understanding of blockchain technology, including its origin, components, types, consensus mechanisms, and cryptocurrency applications
- To explore public blockchain systems and smart contracts, their principles, implementations, and industry applications
- To know the security issues of blockchain technology
- To understand security challenges in blockchain and explore industry applications.
- To study practical blockchain implementations through case studies

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze blockchain technology by examining its origin, components, types, consensus mechanisms, and cryptocurrency applications

**CO-2:** Evaluate public blockchains and smart contracts, their principles, implementations, and industry applications

**CO-3:** Examine private and consortium blockchains, their applications, and the role of ICOs in blockchain fundraising

**CO-4:** Apply blockchain security principles and explore its applications, limitations, and future scope

**CO-5:** Implement blockchain solutions through case studies and development with Python and Hyperledger Fabric

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 2                     | 2    | 2    | 1    | 1    | 1    | 1    | 2    | 1    | 1     | 1     | 2     | 2                               | 2     | 2     | 1     |
| CO-2 | 3                     | 2    | 3    | 3    | 3    | 2    | 2    | 3    | 2    | 1     | 3     | 3     | 3                               | 3     | 2     | 1     |
| CO-3 | 2                     | 2    | 2    | 1    | 1    | 1    | 1    | 2    | 1    | 1     | 1     | 2     | 3                               | 3     | 3     | -     |
| CO-4 | 3                     | 3    | 2    | 1    | 1    | 1    | 1    | 2    | 1    | 1     | 2     | 3     | 3                               | 3     | 2     | 3     |
| CO-5 | 3                     | 3    | 3    | 1    | 1    | 1    | 1    | 1    | 1    | 1     | 1     | 2     | 2                               | 2     | 3     | 2     |

#### UNIT-I:

**Fundamentals of Blockchain:** Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency – Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.

#### UNIT-II:

**Public Blockchain System:** Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts:

Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

#### **UNIT-III:**

**Private Blockchain System:** Introduction, Key Characteristics of Private Blockchain, Why We Need Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.

**Consortium Blockchain:** Introduction, Key Characteristics of Consortium Blockchain, Why We Need Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.

#### **UNIT-IV:**

**Security in Blockchain:** Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric.

**Applications of Blockchain:** Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.

#### **UNIT-V:**

**Blockchain Case Studies:** Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities.

Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

#### **TEXT BOOK:**

1. Block Chain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

#### **REFERENCES:**

1. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
2. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Pearson Addison Wesley

#### **ONLINE RESOURCES:**

1. The Bitcoin Backbone Protocol - Analysis and Applications, J. A. Garay et al., EUROCRYPT2015LNCSVOI9057, Vol. II, pp. 281310 (eprint.iacr.org/2016/1048)

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PC1AM304) NATURAL LANGUAGE PROCESSING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn the fundamentals of natural language processing
- To understand the various text processing techniques in NLP
- To recognize the role of deep learning techniques in text classification within the field of NLP
- To use topic modelling, case studies and apply the NLP techniques to IR applications
- To gain the knowledge of NLP Pipeline in real time applications like health care, E-commerce etc.

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basic concepts of natural language processing pipeline and applications of NLP

**CO-2:** Illustrate various text representation techniques in NLP

**CO-3:** Analyse text classification techniques and deep learning basics to process natural language text

**CO-4:** Demonstrate text summarization methods with an example

**CO-5:** Apply NLP Pipelines to solve real world applications

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 2                     | 2    | 2    | 1    | 1    | 2    | 1    | 1    | 1    | 1     | 1     | 1     | 2                               | 1     | 2     | 1     |
| CO-2 | 2                     | 2    | 2    | 2    | 1    | 2    | 1    | 1    | 1    | 1     | 1     | 1     | 3                               | 2     | 3     | 1     |
| CO-3 | 2                     | 3    | 2    | 2    | 2    | 3    | 1    | 1    | 1    | 1     | 1     | 1     | 3                               | 3     | 2     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 2    | 2    | 1    | 1    | 2    | 1     | 1     | 1     | 2                               | 3     | 2     | 2     |
| CO-5 | 3                     | 2    | 2    | 2    | 3    | 3    | 1    | 2    | 2    | 1     | 1     | 2     | 3                               | 2     | 3     | 2     |

#### UNIT-I:

**NLP - A Primer:** NLP in the Real World, NLP Tasks, NLP Levels, What Is Language, Building Blocks of Language, Why Is NLP Challenging, Machine Learning and Overview Approaches to NLP, Heuristics-Based, Machine Learning, Deep Learning for NLP.

**NLP Pipeline:** Data Acquisition, Pre-Processing Preliminaries Frequent Steps, Advanced Processing Feature Engineering Classical NLP/ML Pipeline DL Pipeline Modelling, Evaluation of Models, Post-Modelling Phases.

#### UNIT-II:

**Text Representation** Vector Space Models, Basic Vectorization Approaches, One-Hot Encoding Bag of Words, Bag of N-Grams, TF-IDF, Distributed Representations, Word Embedding, Going Beyond Words, Distributed Representations.



**UNIT-III:**

**Text Classification:** Applications, One Pipeline, Many Classifiers, Using Neural Embeddings in Text Classification, Deep Learning for Text Classification Interpreting Text Classification Models.

**Deep Learning for Text Classification:** CNNs for Text Classification, LSTMs for Text Classification.

**UNIT-IV:**

**Topic Modelling:** Text Summarization, Use Cases Setting Up a Summarizer: An Example Recommender Systems for Textual Data Machine Translation Question-Answering Systems, Social Media, E-Commerce and Retail, Healthcare, Finance and Law

**UNIT-V:**

**Case Study on NLP Pipeline, Text Classification:** Ticketing, Ecommerce, Social media, health care, Recommender systems and other applications of NLP

**TEXT BOOKS:**

1. Practical Natural Language Processing: A Comprehensive Guide to Building Real world NLP Systems, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta & Harshit Surana, 1<sup>st</sup> Edition, O'Reilly Media, 2020
2. Natural Language Understanding, James Allen, Benjamin Cummings, 2<sup>nd</sup> Edition, 1995

**REFERENCE:**

1. Natural Language Processing and Information Retrieval, Tanveer Siddiqui, U. S. Tiwary, Oxford University Press, 2008

**ONLINE RESOURCES:**

1. <https://nptel.ac.in/courses/106101007/>
2. <http://www.cs.colorado.edu/~martin/sp2.html>
3. <https://web.stanford.edu/~jurafsky/sp3/>

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1CS405) SOCIAL MEDIA ANALYTICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Graph Theory, Data Mining, Python/R programming

#### COURSE OBJECTIVES:

- To familiarize with the concept of social media
- To familiarize with the concept of social media analytics and understand its significance
- To enable to develop skills required for analyzing the effectiveness of social media
- To examine the ethical and legal implications of leveraging social media data
- To familiarize with different visualization techniques for Social media analytics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the concept of social media Analytics and its significance

**CO-2:** Analyze the effectiveness of social media

**CO-3:** Use different effective Visualization techniques to represent social media analytics

**CO-4:** Use different Social media analytics tools effectively and efficiently

**CO-5:** Acquire the fundamental perspectives and hands-on skills needed to work with social media data

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 3                     | 3    | 2    | 3    | 3    | 2    | -    | -    | 2    | 2     | 2     | 3     | 3                               | 3     | 3     | 3     |
| CO-2 | 3                     | 2    | 3    | 2    | 3    | 3    | -    | -    | 2    | 2     | 2     | 3     | 3                               | 3     | 3     | 3     |
| CO-3 | 3                     | 2    | 3    | 3    | 3    | 3    | -    | -    | 3    | 2     | 3     | 2     | 3                               | 3     | 3     | 3     |
| CO-4 | 3                     | 2    | 2    | 3    | 2    | 2    | -    | -    | 3    | 2     | 2     | 3     | 3                               | 2     | 2     | 2     |
| CO-5 | 3                     | 2    | 2    | 3    | 2    | 3    | -    | -    | 2    | 2     | 3     | 2     | 3                               | 2     | 3     | 3     |

#### UNIT-I:

**Social Media Analytics:** An Overview: Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations. Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools

#### UNIT-II:

**Social Network Structure, Measures & Visualization:** Basics of Social Network Structure - Nodes, Edges & Tie Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust Network Visualization - Graph Layout, Visualizing Network features, Scale Issues. Social Media Network Analytics - Common

Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.

#### **UNIT-III:**

**Social Media Text, Action & Hyperlink Analytics:** Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text 8 Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media Hyperlink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools

#### **UNIT-IV:**

**Social Media Location & Search Engine Analytics:** Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools

#### **UNIT-V:**

**Social Information Filtering:** Social Information Filtering - Social Sharing and filtering, Automated Recommendation systems, Traditional Vs social Recommendation Systems Understanding SocialMedia and Business Alignment, Social Media KPI, Formulating a Social Media Strategy, Managing Social Media Risks

#### **TEXT BOOKS:**

1. Seven Layers of Social Media Analytics - Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data, Gohar F. Khan, 2015
2. Mining the Social Web - Analyzing Data from Facebook, Twitter, LinkedIn, and Other Social Media Sites, Matthew A. Russell, O'Reilly
3. Social Network Data Analytics, Charu Aggarwal (Ed.), Springer, 2011

#### **REFERENCES:**

1. Analyzing the Social Web, Jennifer Golbeck, 1<sup>st</sup> Edition
2. Social Media Analytics, Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press, 2015
3. Social Media Analytics Strategy - Using Data to Optimize Business Performance, Alex Gonçalves, APress
4. Social Media Data Mining and Analytics, Szabo G., G. Polatkan, O. Boykin & A. Chalkiopoulos, Wiley, 2019
5. Social Media Analytics: Effective Tools for Building, Interpreting and Using Metrics, Marshall Sponder, 1<sup>st</sup> Edition, McGraw-Hill, 2011

#### **ONLINE RESOURCES:**

1. <https://cse.iitkgp.ac.in/~pawang/courses/SC16.html>
2. [https://onlinecourses.nptel.ac.in/noc20\\_cs78/preview](https://onlinecourses.nptel.ac.in/noc20_cs78/preview)

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1DS405) TIME SERIES ANALYSIS AND FORECASTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn basic analysis of time series data and time series regression
- To learn auto-regressive and model averaging models
- To learn basic concepts of forecasts for a time series data
- To learn how to perform the advanced computation like Multivariate Time Series Models and Forecasting

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply ideas to real time series data and interpret outcomes of analyses

**CO-2:** Learn how to construct a time-series plot and identifying the underlying patterns in the data

**CO-3:** Learn basic concepts factorization techniques in time series analysis

**CO-4:** Learn how to develop forecasts for a time series that has a seasonal pattern

**CO-5:** Demonstrate an advanced understanding of the underlying concepts in the time series and frequency domains for computation, visualization, and analysis of time series data

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 3                     | 3    | 2    | 3    | 3    | 2    | -    | -    | 2    | 2     | 2     | 3     | 3                               | 2     | 1     | 2     |
| CO-2 | 3                     | 2    | 3    | 2    | 3    | 3    | -    | -    | 2    | 2     | 2     | 3     | 2                               | 1     | 2     | 2     |
| CO-3 | 3                     | 2    | 3    | 3    | 3    | 3    | -    | -    | 3    | 2     | 3     | 2     | 3                               | 2     | 2     | -     |
| CO-4 | 3                     | 2    | 2    | 3    | 2    | 2    | -    | -    | 3    | 2     | 2     | 3     | 3                               | 2     | 3     | 1     |
| CO-5 | 3                     | 2    | 2    | 3    | 2    | 3    | -    | -    | 2    | 2     | 3     | 2     | 3                               | 2     | 3     | 3     |

#### UNIT-I:

Introduction of time series analysis, The Nature of Time Series Data, Internal structures of time series, Time Series Statistical Models, Moving Window for Time series data.

#### UNIT-II:

Fourier decomposition, deep learning features LSTM, Image Histogram, Deep learning feature CNN, Relational data, Graph data, Indicator Variable, Feature binning, interaction variable.

#### UNIT-III:

Mathematical transforms, Model specific featurizations, Feature orthogonality, Domain specific featurizations, Feature slicing.

Kaggle Winner Solutions: Students are suggested to carry out with at least 3 Kaggle winner competition case study solutions and present the report for that.

(<https://www.kaggle.com/code/sudalairajkumar/winning-solutions-of-kaggle-competitions>)

#### **UNIT-IV:**

Autocorrelation and Partial autocorrelation, General Approach to Time Series Modelling and Forecasting, Evaluating and Monitoring Forecasting Model Performance, Introduction Least Squares Estimation in Linear Regression Models, Statistical Inference in Linear Regression.

#### **UNIT-V:**

Model Adequacy Checking, Regression Models for General Time Series Data. Exponential Smoothing, First order and Second order, Multivariate Time Series Models and Forecasting, Multivariate Stationary Process, ARIMA model, Vector AR (VAR) Models.

#### **TEXT BOOKS:**

1. Introduction to Time Series Analysis and Forecasting, Douglas C. Montgomery, Cheryl L. Jen, 2<sup>nd</sup> Edition, Wiley Series in Probability and Statistics, 2015
2. Master Time Series Data Processing, Visualization, and Modeling Using Python, Dr. Avishek Pal, Dr. P. K. S. Prakash, 2017

#### **REFERENCES:**

1. Introduction to Time Series Analysis, P. J. Brockwell and R. A. Davis
2. Time Series Analysis and Its Applications, Robert H. Shumway and David S. Stoffer
3. Introduction to Statistical Time Series, W. A. Fuller
4. Time Series Analysis, Wilfredo Palma

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1AM302) EXPLAINABLE AI

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce the concepts of explainable AI and interpretable machine learning
- To understand concepts of black-box machine learning, explainable AI methods
- To analyse various explainable AI methods
- To understand various applications of explainable deep learning
- To identify various challenges related to explainable AI

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the context of the machine learning applications with respect to explain ability and interpretability

**CO-2:** Analyze concepts of various ML model validation, evaluation, and performance visualization for both supervised and unsupervised learning

**CO-3:** Apply state of the art explainable AI tools, techniques and interpretable methods for various AI applications

**CO-4:** Correlate explainable deep learning techniques and its applications

**CO-5:** Interpret various XAI applications and future challenges related to Explainable AI

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | -                     | 2    | 2    | 2    | 1    | -    | -    | 1    | -    | -     | -     | -     | 2                               | 1     | 1     | 1     |
| CO-2 | 2                     | -    | 2    | 1    | 2    | -    | -    | 2    | -    | -     | -     | -     | 2                               | 2     | 2     | 1     |
| CO-3 | -                     | 2    | 3    | 2    | 2    | -    | -    | 1    | -    | -     | -     | -     | 2                               | 3     | 2     | 2     |
| CO-4 | -                     | -    | -    | 1    | -    | -    | -    | 2    | -    | -     | -     | -     | 2                               | 3     | 2     | 2     |
| CO-5 | 2                     | -    | 3    | 1    | 1    | -    | -    | 1    | -    | -     | -     | -     | 2                               | 2     | 2     | 2     |

#### UNIT-I:

**Introduction:** Black-Box problem, Goals, Brief History, Purpose, Societal Impact, Types of Explanations, Trade-offs, Taxonomy, Flowchart for Interpretable and Explainable Techniques.

**Pre-model Interpretability and Explainability:** Data Science Process and EDA, Exploratory Data Analysis, Feature Engineering.

#### UNIT-II:

**Model Visualization Techniques and Traditional Interpretable Algorithms:** Model Validation, Evaluation, and Hyperparameters, Model Selection and Visualization, Classification Model Visualization, Regression Model Visualization, Clustering Model Visualization, Interpretable Machine Learning Properties, Traditional Interpretable Algorithms

**UNIT-III:**

**Model Interpretability:** Advances in Interpretable Machine Learning: Interpretable vs. Explainable. Algorithms, Tools and Libraries, Ensemble-Based, Decision Tree-Based, Rule-Based Techniques, Scoring System.

**Post-Hoc Interpretability and Explanations:** Tools and Libraries, Visual Explanation, Feature Importance, Example-Based.

**UNIT-IV:**

**Explainable Deep Learning:** Applications, Tools and Libraries, Intrinsic, Perturbation, Gradient / Backpropagation.

**UNIT-V:**

**Explainability:** Time Series Forecasting, Natural Language Processing, and Computer Vision XAI: Challenges: Properties of Explanation, Categories of Explanation, Taxonomy of Explanation Evaluation.

**XAI: Future:** Formalization of Explanation Techniques and Evaluations, Adoption of Interpretable Techniques, Human-Machine Collaboration, Collective Intelligence from Multiple Disciplines, Responsible AI (RAI), XAI and Security, Causality and XAI.

**TEXT BOOK:**

1. Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning, Uday Kamath and John Liu, 1<sup>st</sup> Edition, Springer, 2021

**REFERENCES:**

1. Explainable AI with Python, Leonida Gianfagna and Antonio Di Cecco, 1<sup>st</sup> Edition, Springer International, 2021
2. Hands-On Explainable AI (XAI) with Python, Denis Rothman, 1<sup>st</sup> Edition, Packt Publishing, 2020

**B.Tech. VIII Semester**

**(22PE1DS406) QUANTUM COMPUTING**

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE OBJECTIVES:**

- To understand the basics of quantum computation and quantum mechanics
- To apply different models of quantum computation
- To illustrate the quantum models using machine learning algorithms
- To explore various Quantum algorithms

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explore the quantum computation environment

**CO-2:** Identify the implications of quantum computation in the field of mathematics and associated fields

**CO-3:** Demonstrate building blocks and programming methodologies for quantum computing

**CO-4:** Apply quantum computation algorithms

**COURSE ARTICULATION MATRIX:**

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 | PSO-3 | PSO-4 |
| CO-1 | 3                     | 2    | 2    | 2    | 3    | 2    | -    | -    | 2    | 2     | 2     | 3     | 1                               | -     | 1     | -     |
| CO-2 | 2                     | 2    | 3    | 2    | 3    | 3    | -    | -    | 2    | 2     | 2     | 3     | 2                               | -     | 1     | 1     |
| CO-3 | 2                     | 2    | 3    | 3    | 3    | 3    | -    | -    | 3    | 2     | 3     | 2     | 1                               | 1     | 1     | 1     |
| CO-4 | 3                     | 2    | 2    | 3    | 2    | 2    | -    | -    | 3    | 2     | 2     | 3     | 1                               | -     | 1     | 2     |

**UNIT-I:**

**Introduction and Background:** Overview, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation. A Preview of Quantum Physics.

**Linear Algebra and The Dirac Notation:** The Dirac Notation and Hilbert, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators

**UNIT-II:**

**Qubits and The Framework of Quantum Mechanics:** The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems.

A Quantum Model of Computation: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations.

**UNIT-III:**

**Superdense Coding and Quantum Teleportation:** Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.



Quantum Error Correction: Error Models for Quantum Computing, Encoding, Error Recovery. Architecture: Bits and Qubits, Classical Gates, Reversible Gates, Quantum Gates.

#### **UNIT-IV:**

**Programming Languages:** Programming in a Quantum World, Quantum Assembly Programming, Toward Higher-Level Quantum Programming.

**Theoretical Computer Science:** Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations.

#### **UNIT-V:**

**Quantum Algorithms:** Deutsch's Algorithm, The Deutsch–Jozsa Algorithm, Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm

#### **TEXT BOOKS:**

1. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press
2. Quantum Computing for Computer Scientists, Noson S. Yanofsky, Michael A. Mannucci, Cambridge University Press

#### **REFERENCES:**

1. Quantum Computing, Ronald De Wolf, QuSoft, CWI and University of Amsterdam
2. Quantum Computing Explained, David McMahon, Wiley
3. Quantum Computing Fundamentals, William (Chuck) Easttom